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Automated Prioritization and Routing of IT Support Tickets Using Machine Learning

by

Mohamed Almarzooqi

**A Thesis Submitted in Partial Fulfilment of the Requirements for the
Degree of Master of Science in Professional Studies: Data Analytics**

Department of Graduate Programs & Research

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Abstract

The increasing complexity of IT service management underscores the need for efficient and scalable solutions to manage IT tickets. This thesis explores the application of machine learning techniques for automated prioritization & routing of IT tickets. So by leveraging publicly available dataset from Kaggle containing IT ticket the study aims to reduce manual intervention, improve resolution times, & optimize resource allocation within IT environments.

The study will evaluate various machine learning algorithms, including SVM, Random Forest, & ensemble models, for their ability to classify and prioritize tickets accurately. to accurately classify and prioritize tickets. Advanced data preprocessing techniques like TF-IDF vectorization and class balancing are used to handle data inconsistencies and imbalances. Additionally, the study also explores hybrid approaches that combine machine learning with rule-based systems are investigated to enhance classification performance for low-frequency and ambiguous ticket categories. Moreover, incorporating feedback loops & real-time data updates ensures model adaptability to evolving IT environments.

The expected outcomes include significant improvements in classification accuracy and the development of a scalable framework for real-time IT ticket management. By using a publicly available dataset, this research aims to provide a framework for organizations looking to enhance service efficiency and comply service-level agreements standards.

Keywords: IT ticket management, machine learning, automated prioritization, ticket routing, hybrid models, SVM, IT service management (ITSM).

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Chapter 1

1.1 Introduction

The growth of information technology (IT) systems in modern organizations has greatly brought significant advancements in operational efficiency, but it has also introduced challenges in managing IT incident and service requests effectively. IT Service Management (ITSM) frameworks, such as those guided by the IT Infrastructure Library (ITIL), aim to streamline and improve processes, as manual handling of IT service tickets remains a major bottleneck. The manual classification, prioritization, and routing of tickets often lead to delays, errors, and inconsistent resolution quality, negatively impacting customer satisfaction and organizational performance.

Machine learning (ML) and artificial intelligence (AI) offer promising solutions to automate IT ticket management. Machine learning models can process large volumes of historical ticket data to automate the categorization, prioritization, and routing of IT service tickets. These models reduce human intervention, leading to faster response times and more efficient resource allocation.

This thesis proposes to design an automated system for IT ticket management by leveraging machine learning. This system will incorporate advanced classification algorithms, natural language processing (NLP), and other methods to create a flexible framework that can adapt to changing IT environments. The goal is to reduce operational overhead, ensure compliance with service-level agreements (SLAs) and improve the overall quality of IT services.

1.2 Project Goals

The main goal of this research is to design and evaluate an automated IT ticket management system that uses machine learning for prioritization and routing. The key objectives include:

- Automates the classification and routing of IT service tickets into appropriate resolution categories.
- Implementing prioritization mechanisms to handle high-impact or urgent tickets more efficiently.
- Improving the accuracy of ticket assignments to minimize the effort of rerouting the tickets and misclassifications.
- Reducing overall ticket resolution times to enhance operational efficiency and user satisfaction.
- Creating a scalable and adaptable framework capable of handling evolving ticket categories and increasing data volumes.

This system is intended to tackle common organizational challenges such as ticket overload, limited staff availability, and the need for immediate decision-making.

1.3 Aims and Objectives

The aims and objectives of this research are as follows:

1. Investigate and compare various machine learning techniques for classifying and prioritizing IT service tickets, including Support Vector Machine, Random Forest, and ensemble methods.
2. Implement natural language processing (NLP) techniques such as TF-IDF vectorization and feature extraction to process unstructured ticket descriptions effectively.
3. Develop a hybrid approach that uses machine learning with rule-based systems to address ambiguous and low-frequency ticket categories.
4. Assess and evaluate the proposed system's effectiveness using real-world datasets with metrics such as classification accuracy, prioritization efficiency, and resolution time reduction.
5. Explore mechanisms for continuously update the system with new data to adapt to evolving IT environments and maintain optimal performance.

1.4 Research Methodology

To achieve the outlined goals, this research follows a structured methodology:

1. **Data Collection:** Collect historical IT ticket data from Kaggle, including details like ticket descriptions, categories, priorities, timestamps, and resolution times.
2. **Data Preprocessing:** involves applying preprocessing techniques such as text cleaning and followed by text normalization through stemming and lemmatization to reduce words to their base forms. The textual data is then transformed into numerical formats using vectorization techniques like TF-IDF or word embeddings suitable for ML. Missing data will be addressed through imputation to ensure data completeness.
3. **Algorithm Selection:** Explore and evaluate various machine learning models, including supervised learning algorithms SVM, Random Forest, Gradient Boosting, and ensemble techniques to find the best performing one.
4. **Hybrid Model Development:** Design a hybrid system that uses machine learning with rule-based approaches to handle complex or unclear tickets.
5. **Performance Evaluation:** Measure the effectiveness of the system using metrics such as accuracy, precision, recall, F1 score, resolution time, and workload distribution.
6. **Iteration and Optimization:** refining the models based on feedback from performance evaluations and explore advanced techniques such as ensemble learning and deep learning to further enhance the system..

This methodical approach ensures a comprehensive analysis of the potential and limitations of machine learning in IT ticket management.

1.5 Limitations of the Study

While this research presents a promising framework for automating IT ticket management, several limitations must be acknowledged:

- **Data Constraints:** The effectiveness of the machine learning models depends heavily on the quality, quantity, and diversity of the datasets used. Limited or imbalanced datasets can hinder the model's performance.
- **Dynamic Ticket Categories:** IT environments are dynamic and frequently changing, with new categories emerging over time. The system must incorporate mechanisms for continuous learning and updating to stay relevant.
- **Scalability Issues:** Deploying machine learning models in real-time IT environments requires significant computational resources, which may pose challenges for smaller organizations or those with legacy systems.
- **Class Imbalance:** Certain ticket categories may have fewer examples, potentially biasing the model's predictions. Techniques such as oversampling like SMOTE or synthetic data creation may mitigate this issue but require further research.
- **Integration and Adoption:** The integration of the proposed system into existing ITSM workflows may face resistance due to organizational inertia or compatibility issues with legacy systems.
- **Adoption Challenges:** Deploying machine learning models in real-time IT might face challenges due to organizational habits or problems with legacy systems.

Despite these limitations, this study provides a foundation for future research and practical implementation of automated IT ticket management systems.

Chapter 2 – Literature Review

2.1 Literature Review: Automated Solutions for ITSM

In 2021, researchers Feras Al-Hawari and Hala Barham at the German Jordanian University developed a machine learning model using SVM to optimize IT service tickets assignments. They analyzed 1,254 past tickets categorized into 13 service types, applying preprocessing steps like noise removal, stemming, and generating TF-IDF vectors to convert descriptions into numerical data. By integrating additional metadata such as user comments, they boosted the model's accuracy from 59.5% to 81.4%. This approach optimized with Sequential Minimal Optimization, reduced manual ticket handling, decreased technician workloads, improved resolution times and service quality. The model was evaluated using metrics like accuracy, ticket closure rates, and average resolution times, highlighting the necessity for regular updates to maintain performance and effective IT resource allocation. Although the model enhanced response times and met service agreements, the study noted challenges with SVMs handling class imbalances in larger datasets & recommended exploring deep learning or ensemble methods for better accuracy and efficiency in future research.

Sever Iscen and M. Zahid Gürbüz (2019) analyzed machine learning algorithms for IT incident management using WEKA to test classifiers like SVM, Naïve Bayes, k-Nearest Neighbors (kNN), and J48 Decision Trees. They worked with three datasets from a healthcare IT provider. Dataset A with 34,006 tickets, Dataset B with 96,586 tickets, and Dataset C which is a refined version of Dataset A with low-accuracy tickets removed. The preprocessing included TF-IDF vectorization and text normalization. The study showed that SVM outperformed other classifiers like Naïve Bayes and kNN, especially with complex and high-dimensional datasets, achieving 89.7% accuracy on Dataset C. Refinement of datasets by removing ambiguous classes improved SVM's performance by about 10%, emphasizing the importance of data quality. However, SVM required longer training times compared to other classifiers. Research concluded that SVM is the most effective for complex, high-dimensional, and imbalanced IT incident datasets suggesting future research could explore hybrid or ensemble models to enhance classification accuracy & efficiency.

Yasmen Wahba, Nazim H. Madhavji, and John Steinbacher (2022) tackled the challenge of overlapping categories and evolving ticket types in IT support ticket classification. They developed the Hybrid Online Offline Model (HOOM), which merges Hierarchical SVM (HSVM) with Passive Aggressive Classifiers (PAC). The Hybrid Online Offline Model adapts to changing ticket features by using HSVM for precise categorization and PAC for real-time model updates. The researchers evaluated 194,488 tickets across 12 main and

110 subcategories. HSVM employed a strategy of training individual classifiers for each parent category to improve accuracy. PAC adjusted model weights upon detecting changes in data patterns through a chi-square technique, allowing HOOM to handle both historical and new data effectively. The analysis showed HOOM achieved 87% accuracy and F1 score of 78%, outperforming single-algorithms approach. This study highlights the success of combining hierarchical and adaptive methods for robust IT ticket classification.

In 2022, Sebastián Moreno, Wilfredo Yushimito, and Sebastián Hughes developed a Stacked Generalization Basic Ensemble Model (SGBEM) to automate ticket assignment in healthcare IT service environments., this model integrates multiple classifiers to mimic expert routing decisions by using a two-layer architecture. The first layer of SGBEM includes three base classifiers: Logistic Regression, Decision Trees, and Naïve Bayes, each tasked to predict for specific ticket types. The outputs from the first layer are then combined in the second layer using Logistic Regression to reduce variance and improve accuracy. The dataset comprised 1,051 tickets from a Chilean hospital software provider and was evaluated through k-fold cross-validation. Key performance metrics included accuracy, precision, and F1 score. The ensemble model achieved 67% Top-1 accuracy and F1 score of 52.2%, correctly identifying the resolution group within its top three predictions for over 80% of tickets. This approach highlights the benefits of ensemble methods in ITSM by offering a scalable and more reliable solution for automated ticket assignment, significantly reducing errors compared to manual routing.

Salman Ahmed, Muskaan Singh, Brendan Doherty, Effirul Ramlan, Kathryn Harkin, and Damien Coyle (2022) explore the application of Artificial Intelligence for IT Operations (AIOps) to predict IT incident risks. Their analysis covers both machine learning (ML) and deep learning (DL) methods for incident classification, risk forecasting, and proactive system monitoring. Techniques like SVM, Naïve Bayes, and Random Forest are used for classifying structured ticket data, while CNNs, LSTMs, and BERT transformers address unstructured ticket data. Key challenges include class imbalances, limited labeled data, and the resource-intensive nature of DL models. To mitigate these issues, the study emphasizes advanced sampling methods like SMOTE and evaluation metrics like AUROC to improve performance. DL models excel at extracting semantic insights from ticket descriptions but face adoption challenges in resource-constrained environments. The authors advocate for integrating BERT into ITSM workflows to balance computational demands with predictive accuracy. The review concludes that while ML techniques like SVM remain effective, the future lies in developing hybrid models combining ML and DL to address current limitations.

In 2018, Qing Wang et al. developed AISTAR, an intelligent system that recommends automation scripts for resolving IT tickets. AISTAR addresses challenges like cold-starts, evolving categories, and script dependencies by integrating a Hierarchical Multi-Armed Bandit (HMAB) framework with Interactive Collaborative Topic Regression (ICTR). Using

116,429 historical tickets from IBM Tivoli Monitoring, the system preprocesses data through cleaning, tokenization, and clustering. HMAB employs Thompson Sampling and LinUCB to incorporate contextual features and script success rates, while ICTR uses collaborative filtering to group similar tickets. AISTAR achieved an 87% Relative Success Rate, outperforming baseline methods and effectively handling cold-start scenarios. Its scalable ensemble approach efficiently manages large datasets and reduces manual workload for IT analysts. The study concludes that AISTAR enhances IT ticket management and plans to integrate advanced NLP & multilingual support in future work.

In 2018, Ruanda Qamili, Shaban Shabani, and Johannes Schneider introduced a machine learning framework for IT service management, automating tasks like ticket categorization, sentiment analysis, and spam detection. The framework tackled issues like noisy data and unbalanced categories by analyzing 18,917 ticket descriptions from a Liechtenstein software company, excluding fields like timestamps. Its three main features included: Spam Detection, which combined SVM, Naïve Bayes, and Random Forest to achieve 99.85% accuracy; Automatic Ticket Assignment, using SVM and Random Forest with 86% accuracy; and Sentiment Analysis, leveraging pretrained NLP tools to prioritize negative sentiments for better customer satisfaction. Metrics like accuracy, precision, recall, and F1-score were used, with SVM being the most consistent. The study emphasized balancing training data to reduce bias and recommended oversampling and data correction. Future improvements included exploring deep learning and advanced ensemble models for greater accuracy and scalability. This research demonstrates the potential of machine learning to optimize IT service operations.

In 2018, Patrick Kubiak and Stefan Rass examined data-driven methods in ITSM, focusing on ticket classification, event mining, log analysis, and proactive monitoring. They identified four ITSM stages: monitoring, alerting, ticketing, and resolution. The study showed that pattern mining and clustering helps detect anomalies and dependencies in log data. Techniques like Sequential Pattern Mining and frequent pattern help identify frequent event sequences that lead to system failures. For ticket classification, Hierarchical Multi-label Classification was used to recognize parent-child relationships in ticket categories, improving routing accuracy. The authors also evaluated models like SVM and Random Forest for classification tasks, and found them suitable for smaller systems. Proactive monitoring was addressed through predictive maintenance methods like Online Failure Prediction and anomaly detection to anticipate system problems. To assess imbalanced dataset's performance, they recommended metrics like AUROC & Hamming Loss. This review tackled real-world ITSM challenges like scalability, data diversity, & uneven ticket distributions. It proposed integrating proactive & reactive strategies for more reliable IT systems & suggested exploring reinforcement learning & NLP for flexible ticketing solutions in future studies.

In 2021, V. V. Nikulin, S. D. Shibaikin, and A. N. Vishnyakov explored using machine learning to automate IT service ticket classification and routing within the ITIL framework. Testing models like CNNs, Gradient Boosting (XGBoost), and Simple Neural Networks on a 10,000-ticket dataset categorized into three hierarchical levels, they used TF-IDF vectorization and chi-square dimensionality reduction during preprocessing. XGBoost emerged as the most effective model, achieving 95% accuracy, surpassing CNN (92%) and Simple Neural Networks (91%), due to its handling of imbalanced datasets and complex feature interactions. The study, leveraging accuracy, F1 scores, and confusion matrices, demonstrated XGBoost's superiority in managing hierarchical ticket structures. It emphasized the benefits of data-driven approaches over rule-based methods, improving ticket routing, reducing resolution times, and optimizing IT resources. The authors suggested future research on combining Gradient Boosting with CNN and domain-specific embeddings to enhance ITIL-aligned ticket management.

S. P. Paramesh and K. S. Shreedhara (2019) created a machine learning-based automated IT service desk system to classify and route problem tickets. They addressed challenges such as misclassification, noisy data, and imbalanced ticket categories using a dataset of 10,742 real-world tickets. Preprocessing steps like data with stop word removal and TF-IDF vectorization, and chi-square tests for feature selection. They compared with Four ML models Multinomial Naïve Bayes, Logistic Regression, k-Nearest Neighbors, and SVM were evaluated. SVM achieved the highest accuracy of 89%. Techniques like oversampling and manual relabeling improved performance on imbalanced data. The automated system successfully reduced ticket resolution times and misclassification rates, enhancing operational efficiency and customer satisfaction. The authors suggested future research on ensemble models, advanced deep learning methods, and more sophisticated feature extraction to further improve automated ticket handling in IT environments.

In 2020, Shivali Agarwal and colleagues developed an AI-driven system to automate helpdesk email ticket assignments, handling over 100,000 tickets monthly with a 90% coverage rate. The hybrid approach combines machine learning classifiers like SVM and Multilayer Perceptron with a Siamese LSTM network for rare categories. Data preprocessing included noise reduction, stemming, and TF-IDF vectorization. A rule-based engine managed low-frequency or ambiguous tickets, while a sliding-window retraining and active learning enhanced accuracy. The system achieved 90% assignment accuracy, saving over 50,000 manual hours annually and improving precision and recall for infrequent categories. Future work may explore deep learning and ensemble methods to further enhance effectiveness.

Atri Mandal, Nikhil Malhotra, Shivali Agarwal, Anupama Ray, and Giriprasad Sridhara (2019) introduced an AI-based system to improve IT helpdesk ticket classification and routing, aiming to enhance accuracy, reduce misclassification, and speed up resolutions.

Their system combines machine learning classifiers and rule-based methods to efficiently assign tickets to resolver groups, boosting IT service quality. Analyzing over 700,000 tickets from various clients, they achieved near-human classification accuracy. Data preprocessing involved text cleaning, tokenization, stemming, and feature extraction using TF-IDF and GloVe embeddings. Frequent categories were handled by SVM and MLP classifiers, while complex cases used a CNN-LSTM hybrid model. A rule engine applied domain-specific rules for ambiguous cases, and a sliding-window retraining approach ensured the system stayed up-to-date. The system reached 90% accuracy and 93% coverage, with the CNN-LSTM model excelling in rare cases and the rule engine managing edge scenarios. This research highlights how combining machine learning and rule-based systems can automate large-scale IT helpdesk operations effectively. Future work includes exploring active learning and transformer-based models.

In 2022, Fahad ul Hassan, Tuyen Le, Chau Le, and K. Joseph Shrestha proposed a hybrid system merging machine learning (ML) and fuzzy logic to automate the prioritization of construction project requirements. Using fuzzy logic to assess risks based on severity, probability, and detectability, the study processed 1,331 real-world requirements. Gaussian membership functions yielded the most balanced risk scores, generating 42 values for ML classification. After preprocessing (text cleaning, tokenization, lemmatization, and vectorization), four ML models were evaluated—Naïve Bayes, SVM, Logistic Regression, and Random Forest. Random Forest outperformed the others with a precision of 78.18%, recall of 77.75%, and an F1-score of 75.82%. TF vectorization proved more effective than TF-IDF for the dataset's small vocabulary. The study demonstrates how fuzzy logic and ML integration addresses expert judgment ambiguities and enhances dataset balance. Future work may focus on deep learning and real-time prioritization to improve scalability and expand applications, including IT ticket management.

This research provides key guidance for IT service providers aiming to automate and scale incident prioritization and management. Cristina Kadar et al. (2011) propose an automated approach to classify IT change requests, reducing errors in manual processes and improving service quality. Since over 60% of major IT issues arise from poorly managed changes, the study emphasizes improving categorization to strengthen risk analysis and resource allocation. The authors analyzed 1,317 tickets from two enterprises (54 and 88 categories) using three methods: information retrieval (IR), supervised learning, and semi-supervised learning. Preprocessing included stemming, lemmatization, and TF-IDF vectorization. Supervised learning with multinomial logistic regression and active learning achieved the best accuracy (88.6% top1, 96.2% top5 for Customer A), while semi-supervised Generalized Expectation reached 74.2% top1 and 88.3% top5 accuracy with minimal labeled data. IR was the least effective, at 53.8% top1 accuracy. The study concludes supervised learning is the most accurate and efficient,

while semi-supervised learning offers a budget-friendly alternative. Future research should consider advanced techniques like transformers and ensembles to boost results further.

Kraidet Khowongprasoe and Taravichet Titijaroonroj (2022) developed an automated system to classify IT tickets in Thai for an IT infrastructure company, aiming to minimize errors from manual categorization. Using a dataset of 360 tickets (2019–2021) with a 53% manual error rate, they classified tickets into incident, customer request, or sales categories. Advanced Thai text tokenizers, including newmm, longest, multicut, and deepcut, were used alongside TF-IDF for feature extraction. Classifiers like SVM, Naïve Bayes, Logistic Regression, and KNN were tested, with SVM achieving the highest accuracy of 82% (3.61% standard deviation). It performed best in the sales category, with a precision of 0.75, recall of 0.89, and F1-score of 0.81, despite class imbalance. In contrast, Naïve Bayes and KNN reached 74% and 71% accuracy, respectively. The study underscores the value of customized tokenization and TF-IDF for Thai text, highlighting SVM's scalability for companies with limited historical data. Future directions could include ensemble methods or neural networks to enhance accuracy. The approach significantly reduces manual errors and improves IT ticket classification consistency.

The 2020 study by Nicolas Ferland and colleagues introduces a hybrid NLP framework to automate IT trouble ticket resolution by combining supervised and unsupervised learning to address challenges like varied descriptions, sparse data, and scalability. It uses historical data and a dataset of 20,000 tickets from the Bitext Customer Support Dataset (27 intents and resolutions), with preprocessing steps such as stop-word removal, lemmatization, and TF-IDF vectorization. The framework combines LDA for clustering themes, LSTM for predicting common resolutions, and Siamese networks for rare cases, unified through a stacking ensemble with logistic regression. This hybrid approach achieved 94.1% top-1 and 95.9% top-5 accuracy, outperforming individual models by 10-20%. The research demonstrates the effectiveness of hybrid NLP models in IT automation and suggests transformer-based models for future scalability and performance improvements.

Rajeev Gupta, K. Hima Prasad, and Mukesh Mohania (2008) present a framework to automate IT Service Management (ITSM) incident management using machine learning and configuration data. Their system enhances incident classification, failure detection, and ticket routing by leveraging data from the Configuration Management Database (CMDB). The approach integrates rule-based systems, NLP techniques for keyword extraction, and an SVM classifier to categorize tickets, with a sliding window mechanism for model updates. Achieving a top-1 accuracy of 84%—outperforming keyword-based methods (50%)—the system reduces unnecessary CMDB searches, improving speed and resource utilization. It demonstrates strong performance across hierarchical ticket categories and benefits from regular model updates. The study highlights future directions

like transformer models and knowledge graphs for handling complex relationships and scaling efficiently. This research showcases how machine learning and NLP can automate IT incident management, reducing manual effort while maintaining high accuracy in dynamic IT environments.

In 2022, M. Venkata Subbarao, Kasukurty Venkata Rao, and Ch. Suresh developed a scalable framework using machine learning (ML) and natural language processing (NLP) to automate IT incident response and ticket management. They analyzed 47,664 ServiceNow ITSM records. Preprocessing steps like data with stop word removal, lemmatization, stemming, and TF-IDF with n-grams. SVM achieved 90% classification accuracy, while Latent Dirichlet Allocation (LDA) reached 96.46% clustering accuracy. While the framework effectively reduced resolution times and manual errors, challenges with imbalanced data and dynamic ticket attributes remain. To further improve the system, they suggest exploring advanced neural networks like BERT. Overall, the study highlights ML and NLP's potential to streamline IT ticket management & enhance incident response.

Paramesh S.P. and Shreedhara K.S. (2019) developed a machine learning model to automate the classification and routing of IT tickets, reducing errors, delays & improving service quality. The system utilized ML models and ensemble methods to optimize the workflow. They analyzed a dataset of 8,468 tickets across 18 categories, including hardware, software, and, network issues. Preprocessing steps like noise reduction, tokenization, stemming, and TF-IDF vectorization were applied, followed by feature selection using Chi-square. Decision Trees achieved 86.9% accuracy were ensemble methods, like Random Forest, outperformed others with 89.3% accuracy. Boosting techniques like Adaboost showed some improvement but were less effective than Random Forest. The results highlighted that Random Forest improved accuracy and handled noisy data well. Applying TF-IDF and Chi-square for feature selection also boosted performance addressing class imbalances. Challenges in the study included dealing with class imbalance and high computational costs. The study successfully integrated machine learning and ensemble methods to create an efficient and scalable system for IT ticket classification and routing & suggested exploring neural networks or hybrid AI approaches in future studies.

Paramesh S.P., Ramya C., and Dr. Shreedhara K.S. (2018) examined ensemble machine learning techniques to classify unstructured IT service desk tickets for large enterprises, aiming to outperform manual and single-model approaches. Using 10,742 tickets spanning 18 categories, they applied preprocessing methods like noise removal, stemming, tokenization, and TF-IDF vectorization, along with data balancing through oversampling and undersampling. Among classifiers tested, Decision Trees achieved 86.9% accuracy, while ensemble methods like Bagged Decision Trees and Random Forest scored 92.04% and 91.81%, respectively. A Voting Classifier with Logistic Regression, KNN, and SVM lagged at 81.15%. The study underscores the effectiveness

of Random Forest and ensemble models in handling unstructured data, benefiting from TF-IDF and balanced datasets. Despite challenges such as computational intensity and complex ticket descriptions, the authors advocate exploring deep learning and hybrid models for improved accuracy and scalability in automating IT ticket classification.

Arief Wibowo and Hariyanto (2023) compare Naïve Bayes (NB) and SVM algorithms for classifying helpdesk tickets to reduce manual effort and improve ticket routing in IT helpdesks. They analyzed 1,327 tickets from a dataset of 1,795, applying preprocessing steps like normalization, tokenization, stop-word removal, and TF-IDF vectorization. Both NB and SVM were evaluated using cross-validation based on accuracy, precision, and recall. Naïve Bayes achieved 82.29% accuracy, 86.84% precision, and 79.09% recall, showing high precision but lower recall. SVM outperformed NB with 85.38% accuracy, 84.27% precision, and 89.47% recall, demonstrating better generalization and handling of complex features, making it more suitable for diverse ticket categories. However, SVM faced challenges such as class imbalance and higher computational demands. The authors suggest exploring ensemble methods or deep learning techniques for future enhancements. The study highlights the importance of choosing the right algorithm and effective data preprocessing to automate ticket classification, leading to improved response times and operational efficiency in IT helpdesk environments.

In 2019, Kuruparan Shanmugalingam and colleagues developed a hybrid automation model for IT support at Millennium I.T E.S.P by combining machine learning (ML) and rule-based methods. They used a dataset of 260,000 emails categorized into 73 types, preprocessing the data with tokenization, stop word removal, OCR for attachments, and TF-IDF vectorization. The model integrated ML techniques like Random Forest, XGBoost, and bidirectional LSTM networks with a rule-based layer using Microsoft LUIS. The bidirectional LSTM achieved 74.6% accuracy, increasing to 85.6% with thresholds, while Random Forest and XGBoost reached 76.5%. Overall, the system automated 81% of ticket processing, with ML handling 61% and rule-based methods managing 20%, significantly reducing manual effort. The study highlighted the model's scalability and flexibility but noted challenges with outlier categories & the computational demands of LSTMs. Future research may explore generative models for low-data classes. This framework offers a scalable & efficient solution for automating IT support ticket handling.

Padej Phomasakha Na Sakolnakorn and Phayung Meesad (2008) explored decision tree models for automating IT ticket assignment in the banking sector to reduce delays and costs. Analyzing 14,440 tickets from a Tivoli CTI system, they used keyword extraction and categorization during preprocessing. Five decision tree algorithms—ID3, J48, NBTree, Random Forest, and REPTree—were tested using WEKA. ID3 and random tree achieved the highest accuracy (93.5%), with ID3 selected for its reliability and ability to handle imbalanced datasets, maintaining a 93.06% accuracy on unseen data. The rules generated by ID3 were stored in a knowledge base, enabling real-time ticket routing,

which significantly reduced errors and optimized resource allocation. While effective for structured environments, the study noted decision trees' limitations with unstructured data and recommended integrating natural language processing (NLP) & ensemble methods to improve performance. This research demonstrated the scalability and efficiency of decision tree-based automation in streamlining IT service processes.

In their 2020 study, Yasmen Wahba, Nazim H. Madhavji, and John Steinbacher examined static word embeddings for classifying IT support tickets, comparing them to traditional TF-IDF methods. The dataset included 1.6 million tickets across 32 categories, with preprocessing steps like tokenization, stop-word removal, and lemmatization. The study evaluated TF-IDF, general-purpose Word2Vec (GN_Word2Vec), and domain-specific Word2Vec (SO_Word2Vec) using Logistic Regression and SVM. TF-IDF paired with SVM delivered the best results, achieving 92% accuracy and an F1-score of 0.92. Domain-specific embeddings showed some promise with 89% accuracy, but general-purpose embeddings offered no significant benefits. The study highlighted the limitations of static embeddings for sparse or imbalanced data and recommended dynamic models like BERT or GPT for better results. TF-IDF remains highly effective for structured datasets, with dynamic approaches showing potential for more challenging data.

In 2022, Simon Fuchs, Clemens Drieschner, and Holger Wittges analyzed machine learning applications in support ticket systems, focusing on classification, escalation prediction, and sentiment analysis across industries like IT and banking. Reviewing 41 studies, they found that models such as SVM and Random Forest achieved 85-92% classification accuracy, while neural networks reached up to 99% with large datasets. Random Forest also performed well with 81-90% accuracy on escalation predictions. Challenges identified included class imbalance, limited labeled data, and high computational costs of deep learning. They recommended exploring advanced deep learning techniques like LSTM and BERT with integrating real-time feedback to improve system adaptability. The study highlighted the effectiveness of traditional models and recommended deep learning and hybrid approaches to enhance scalability and handle complex support ticket systems.

S.P. Paramesh, K.S. Shreedhara (2019) investigates the use of ensemble machine learning techniques to automate IT help desk incident classification. The goal is to improve accuracy and reliability, especially for multi-class datasets. The dataset included 10,742 incidents across 18 categories. Preprocessing steps like stop-word removal, stemming, and feature selection using the chi-square test. The authors compared the performance of individual classifiers (Naïve Bayes and SVM) with ensemble methods like Bagging and Boosting (Adaboost) to enhance classification. The findings indicated that Naïve Bayes had an accuracy of 71%, while SVM outperformed it with 85% accuracy. The ensemble methods further improved results, with Bagging-SVM reaching an accuracy of 87.78% and Adaboost-SVM at 86.15%. These methods particularly

enhanced performance in categories with fewer incidents by addressing class imbalances. The study noted that while ensemble models offered better accuracy, they required higher computational resources, posing challenges for real-time applications. Future directions include integrating NLP techniques for feature extraction and exploring deep learning approaches to handle scalability for larger datasets.

In their 2020 study, Aleksandra Revina, Krisztian Buza, and Vera G. Meister examined simple, interpretable machine learning models for classifying IT tickets, highlighting their practicality in resource-constrained IT environments. The dataset of 28,243 tickets was divided into low, medium, and high complexity levels, and two feature extraction methods were evaluated: TF-IDF and linguistic features (e.g., sentiment lexicons, parts-of-speech, word counts). Tested classifiers included Naïve Bayes, Decision Trees, Logistic Regression, and SVM, alongside semi-supervised techniques like QuickSUCCESS and kNN with self-training. Decision Trees and Naïve Bayes paired with linguistic features achieved 89% accuracy and F1-scores, outperforming more complex models. Although SVM with TF-IDF also performed well, it lacked the interpretability of linguistic features. Semi-supervised methods were faster but less precise, making them more suitable for smaller datasets. Linguistic features proved particularly effective for medium and high-complexity tickets. The study identified issues with ambiguous ticket descriptions and suggested future research on dynamic embeddings (e.g., BERT) or hybrid models combining linguistic and neural network approaches.

Sara Silva, Rubén Pereira, and Ricardo Ribeiro (2018) proposed an automated system for categorizing IT incidents to optimize ticket routing and reduce inefficiencies in IT service management. The study analyzed 10,000 tickets in ten categories, such as software and security issues, and tested three methods: using non-textual data (e.g., severity), textual descriptions with preprocessing, and a mix of both. Models like SVM and KNN were trained with TF-IDF features and evaluated via cross-validation. SVM outperformed KNN, achieving 43% accuracy with non-textual data, 86% with textual inputs, and 89% when combining both, while KNN reached 82% with combined data. The research highlighted textual data's critical role and SVM's superior performance, despite challenges like class imbalance. The authors recommended exploring advanced NLP techniques, such as BERT and hybrid models, to improve contextual understanding. This work underscores the potential of machine learning, particularly SVM, to enhance IT incident categorization accuracy and streamline service management processes.

In 2023, Mohammad Agus Prihandono, Ruki Harwahyu, and Riri Fitri Sari evaluated machine learning (ML) and deep learning (DL) algorithms to prioritize IT incident tickets, aiming to improve IT service management through automation. Manual ticket categorization often leads to inefficiencies, which this study seeks to address. Algorithms tested included Random Forest, SVM, Multilayer Perceptron, Recurrent Neural Networks, Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU). Using 46,606

incident tickets with 25 attributes, the researchers refined the data to seven key features using chi-square tests. Hyperparameter tuning via grid search and cross-validation ensured robust results. Deep learning models outperformed traditional ones, with LSTM achieving the highest accuracy (98.866%), followed by GRU and RNNs. Among traditional models, Random Forest led with 98.069% accuracy, surpassing SVM and Multilayer Perceptron. The study highlighted that deep learning models handle complex data well but require more computational resources, whereas Random Forest provides strong performance with lower computational demands, making it suitable for resource-limited environments. The authors recommend exploring ensemble and hybrid models to further enhance efficiency.

Vikrant Shimpi, Maitreya Natu, Vaishali Sadaphal, Vaishali Kulkarni (2021) presents a method for identifying recurring IT issues by mining trouble tickets, with the aim of automating issue identification to enhance IT service management. The study differentiates between system-generated tickets (SGTs) and user-generated tickets (UGTs), each requiring a distinct approach for effective processing. For SGTs, the authors used clustering techniques based on the Dice coefficient to measure textual similarity, grouping tickets with similar descriptions. For UGTs, they applied a keyword extraction framework that generated n-grams (unigrams, bigrams, and trigrams) a method that pulls out key words and phrases to better handle the diverse ways users report issues. The study achieved an accuracy of 76.16% for mapping SGTs to predefined issues, and 81.63% for grouping UGTs, which outperformed manual classification methods. The results demonstrated that combining clustering techniques with domain-specific keyword extraction can significantly improve issue identification accuracy. Furthermore, using a standardized service catalog for system tickets helped quickly resolve common issues like full disks or memory leaks.

Noor Aklima Harun, Sharin Hazlin Huspi, and Noorminshah A. Iahad (2023) seek to automate question classification in Helpdesk Ticketing Support (HTS) systems to minimize misclassification and reduce ticket transfer delays. They employed machine learning techniques, specifically Naïve Bayes (NB) and SVM, to enhance ticket routing efficiency. The study analyzed 7,278 helpdesk tickets from Universiti Teknologi Malaysia's IT support over three years, using tokenization and TF-IDF for feature extraction. Their two-tier classification framework first categorizes tickets into Department, Guideline, or System, then into specific subcategories based on keywords. Evaluation metrics revealed that SVM achieved 89.7% accuracy, outperforming NB's 82.4%. SVM effectively managed complex data and improved results through TF-IDF processing. Automating classification reduced delays and sped up ticket resolution. The study concludes that SVM is ideal for HTS systems and suggests future research on combining models to further enhance performance and efficiency, demonstrating machine learning's significant impact on helpdesk operations and customer satisfaction.

In 2020, Yixin Diao, Hani Jamjoom, and David Loewenstern introduced xPad, a rule-based classification system for IT service tickets, designed to address limited labeled data. Unlike supervised learning, xPad relies on collaboratively created and refined rules from IT analysts, leveraging expert knowledge to improve accuracy. The study analyzed 20,000 tickets, using inclusive rules to identify relevant issues and exclusive rules to prevent misclassification. Preprocessing included keyword extraction and removal of irrelevant information. xPad increased coverage from 70% to 84.37% and achieved 88% training accuracy and 81% testing accuracy, outperforming Naïve Bayes, which reached only 56% in testing. Rule creation for 10 failure codes was efficient, requiring just 5 hours. The authors highlighted xPad's scalability and adaptability, making it a cost-effective solution for dynamic datasets with limited or changing labels, surpassing traditional machine learning approaches in such contexts.

In 2022, Jian Xu, Hang Zhang, Wubai Zhou, Rouying He, and Tao Li developed a signature-based classification framework to automate the categorization of trouble tickets in IT service management. The framework targets both structured system-generated tickets (SGTs) and unstructured user-generated tickets (UGTs), using tailored algorithms to improve classification accuracy and scalability by creating issue-specific signatures. The research utilized a dataset of over 60,000 SGTs and 40,000 UGTs. Preprocessing included tokenization, domain-specific keyword extraction, and the removal of irrelevant terms. They employed n-grams to enhance textual analysis and introduced two algorithms: PSCA_SGT for structured tickets and PSCA_UGT for unstructured tickets. The results showed that PSCA_SGT achieved a 93.79% F1-score, effectively processing sequential signatures, while PSCA_UGT managed an 81.73% F1-score, adept at handling variations in user descriptions. The study found that specialized data preprocessing increased accuracy by about 10%, and the system could efficiently process over 100,000 tickets. This method surpassed traditional grouping methods like k-means by 20%-25% in accuracy. The authors concluded that their signature-based framework provides a scalable, precise, and adaptable solution for categorizing tickets in large-scale IT service management environments, effectively accommodating both structured and unstructured data.

Li Xiao, Jun Wang, and Klara Nahrstedt (2021) introduce Enhanced Ticket-Based Routing (ETBR), an algorithm that improves traditional Ticket-Based Routing (TBR) for the Delay-Constrained Least-Cost Routing problem in IT services. ETBR optimizes ticket distribution and incorporates historical data to enhance routing efficiency and accuracy, aiming to minimize costs while meeting delay requirements in dynamic, high-volume IT networks. Using a simulated network with over 200 nodes and 1,500 connections, ETBR introduces color-based ticket distribution—classifying tickets by cost or delay optimization—and leverages past routing data to prioritize successful routes and reduce unnecessary probes. Evaluation under various conditions showed that ETBR reduced

average path costs by 17% compared to TBR, maintained nearly 100% success rates, and kept message overhead similar, thereby improving performance without increasing network load. The study concludes that ETBR effectively balances cost and delay, utilizes historical data efficiently, and scales well in dynamic environments. Future research may explore integrating machine learning to further enhance routing accuracy and ticket categorization.

Alessandro Zangari and colleagues (2023) examine advancements in IT service management ticket automation through multi-level classification using Transformer-based NLP models like BERT. They analyzed two hierarchically labeled datasets—20,000 customer complaints and 35,000 bug reports—preprocessing the data by cleaning and converting text into BERT embeddings using methods such as averaging and selecting maximum values across layers. The study utilized Multi-Level BERT (ML-BERT) to incorporate hierarchical context, achieving a 5.7% higher F1-score and a 5.4% increase in accuracy compared to baseline and traditional classifiers. Results highlight the significance of hierarchical label structures and advanced embedding strategies, with some approaches improving F1-scores by over 28%. The authors conclude that ML-BERT effectively handles complex multi-level ticket classification and suggest future research on ensemble techniques to further boost performance by combining hierarchical classifiers with other machine learning models.

In 2021, Jian Xu, Rouying He, Wubai Zhou, and Tao Li developed three models—Group Profile (GP), Transfer Profile (TP), and Routing Network (RN)—to improve the efficiency of ticket routing and resolution across 40,000 IT service tickets managed by 150 expert groups. The GP model evaluates the success rates of each group with specific problem types to optimize ticket assignment. The TP model analyzes transfer patterns between groups to identify common paths for unresolved tickets and calculate optimal routing decisions. The RN model combines the insights from GP and TP into a directed graph, integrating expertise and collaboration data to balance skills and inter-group transfers effectively. Performance was assessed by metrics like mean steps to resolve (MSTR) and resolution success rate (RSR). The RN model significantly outperformed traditional methods, achieving a 97% RSR and reducing MSTR by 15% in complex cases. The study concludes that the RN model greatly enhances the ticket resolution process by effectively integrating group performance data and transfer patterns, making it a robust solution for managing IT service tickets.

Jian Xu, Liang Tang, and Tao Li (2023) introduce the System Situation Ticket Identification Ensemble (STI-E) framework to classify situation tickets in IT environments with highly imbalanced datasets, where such tickets represent only 1% of 100,000 total. STI-E employs an ensemble of SVM to enhance accuracy and address class imbalance. The team performed data cleaning, tokenization, and extracted domain-specific keywords, using Synthetic Minority Over-sampling Technique (SMOTE) to balance the

dataset. STI-E consists of multiple SVM classifiers trained on balanced subsets created through SMOTE, selective labeling of high-impact tickets, and diverse sampling to ensure varied training data. Predictions are combined using majority voting, improving performance. The framework achieved 92% precision, 88% recall, and an F1-score of 0.89, outperforming single SVMs and keyword matching by 10-15%. STI-E is scalable and reduces manual effort for IT analysts. The authors conclude that STI-E effectively identifies situation tickets in imbalanced datasets and plan to explore deep learning and real-time classification in future work.-time ticket classification to improve the system's responsiveness and effectiveness.

2.2 Key Takeaways

1. The effectiveness of Utilizing SVM in automating ticket classification, especially for complex IT service management (ITSM) datasets.
2. Importance of Feature Engineering (e.g., dimensionality reduction) are critical for improving model performance in ticket classification and Data Preprocessing (e.g., noise removal, stemming, TF-IDF vectorization)
3. Handling Imbalanced data using techniques like Synthetic Minority Over-sampling Technique (SMOTE) and class weighting that are essential for improving classifier performance on imbalanced datasets.
4. Enhancing computational speed and accuracy by applying techniques like TruncatedSVD allows for effective management of high-dimensional data, making models faster to train while maintaining accuracy by reducing the risk of overfitting.
5. Comprehensive model evaluation using multiple metrics such as precision, recall, F1-score, AUROC, and Top-K accuracy is critical in multi-class classification tasks to ensure that the model performs well across all categories.

Chapter 3- Project Description

3.1 Project Understanding and Goals

This project aims to develop an automated IT ticket classification system to enhance IT Service Management (ITSM) efficiency, following the CRISP-DM framework (Figure 1 CRISP-DM Methodology) through multiple phases.

In business understanding phase, the project focuses on classifying tickets into predefined categories and urgency levels using a Kaggle dataset. The goal is to enhance efficiency by reducing response delays and optimizing resource allocation. Challenges like data imbalance, complex ticket structures, and scalability are addressed to create a reliable solution for modern IT operations.

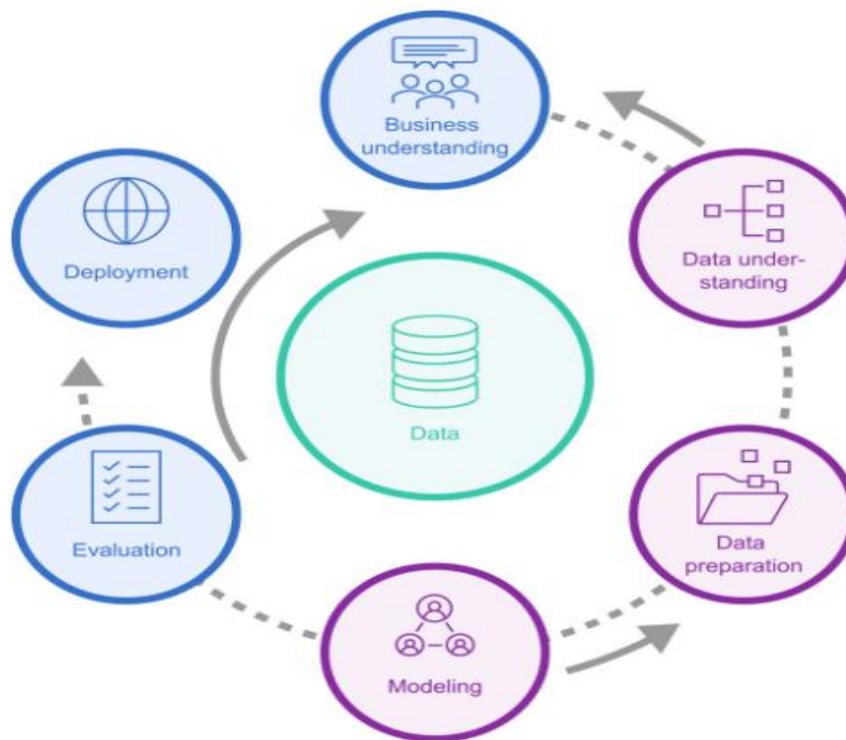


Figure 1 CRISP-DM Methodology

3.2 Data Understanding & Data Preparation

Ticket data from Kaggle is collected and assessed for machine learning applications. The process includes analyzing the dataset's structure, identifying relevant features, and ensuring data quality. Preprocessing involves text cleaning (eliminating noise like special characters and unnecessary text), stemming (standardizing words to root forms), and TF-IDF vectorization (converting text into numerical data for model training).

3.3 Modeling

Once preprocessing is complete, models like SVM, Random Forest, and Gradient Boosting are evaluated to find the most effective ticket classification approach. Steps are taken to identify and mitigate risks, including privacy concerns and biases, ensuring ethical model performance.

3.4 Evaluation

Performance evaluation is vital in the CRISP-DM process, using metrics like accuracy, precision, recall, and F1-score to measure success. Optimization focuses on model tuning and ensemble methods to enhance results. The final output is a scalable, adaptable ticket classification system tailored to ITSM needs, with thorough documentation to support maintenance and future improvements.

Chapter 4- Analysis

4.1 Descriptive Analysis

4.1.1 Introduction

In the field of IT particularly at the level of first-line support, mastering IT ticket prioritization and routing is essential for evaluating the operational and organizational factors influencing resolution efficiency within IT service management. The study incorporates machine learning models like Support Vector Machine, Random Forest, and Gradient Boost to classify and prioritize IT service tickets based on their urgency. **(Error! Reference source not found.** below demonstrates the IT ticket processing workflow, highlighting the integration of these machine learning algorithms into the ticket classification and prioritization process.

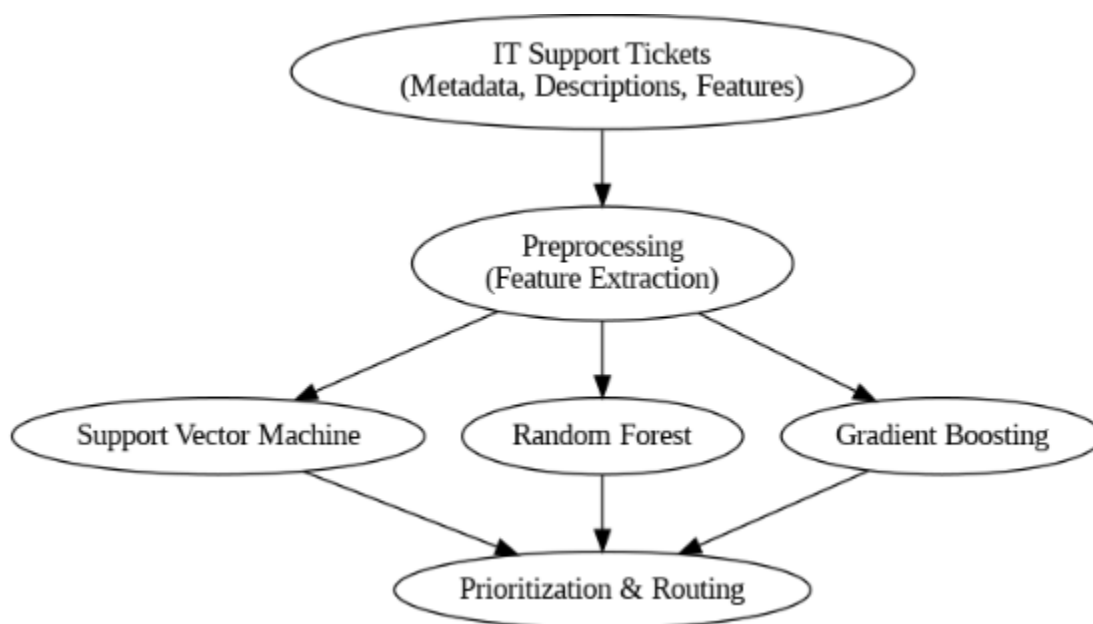
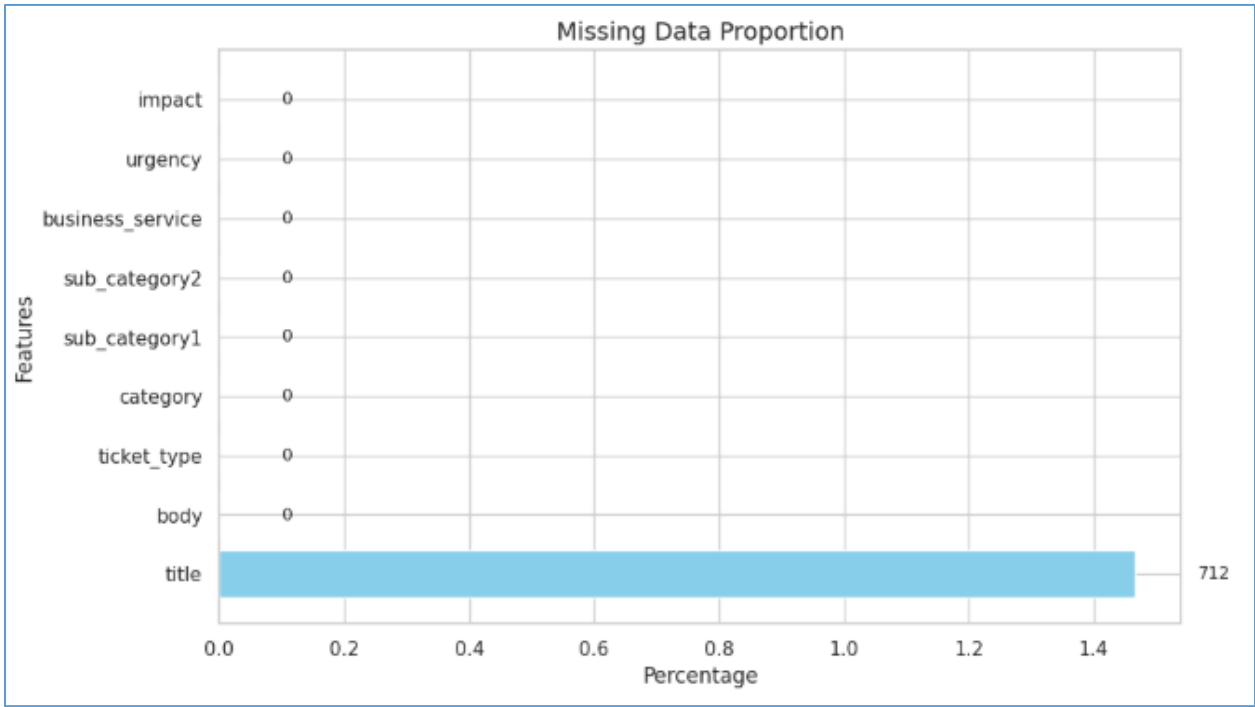


Figure 2 Flow Diagram

4.1.2 Data Understanding

This analysis uses a dataset of 48,549 entries, featuring variables such as ticket metadata, textual descriptions, and categorical data. The dataset, obtained from Kaggle, captures IT support tickets recorded within a defined period. There are 712 missing values in the title field (Figure 3 Missing ValueError! Reference source not found., requiring careful attention during subsequent steps of data preparation.

Figure 3 Missing Value



This dataset integrates textual and categorical fields to deliver meaningful insights into ticket characteristics, urgency, and routing requirements. Textual features like title and body capture detailed contextual information about the tickets, playing a key role in automated classification and urgency determination. Conversely, categorical fields such as ticket_type, category, sub_category1, and sub_category2 provide a structured hierarchy to inform routing processes. Moreover, the business_service attribute associates tickets with appropriate service domains, enabling efficient resource distribution. The unique values in categorical fields, including 13 distinct entries in category and 59 in sub_category1, reflect the dataset's ability to encompass a wide variety of issues.

Table 1 Key Variables Summary provides summary of the key variables in the dataset, including their descriptions, data types, the number of unique values, & the percentage of missing data

Table 1 Key Variables Summary

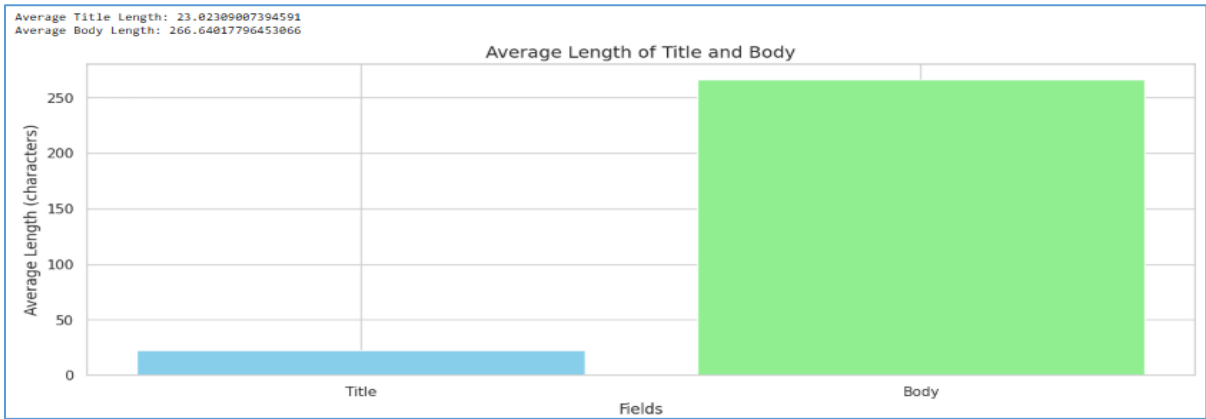
	Variable	Description	Type	Unique Values	Missing %
0	title	Title of the ticket (short description)	object	30118	1.46656
1	body	Detailed description of the issue in the ticket	object	48549	0.00000
2	ticket_type	Type of ticket (incident or service request)	int64	2	0.00000
3	category	Broad category of the ticket (13 categories)	int64	13	0.00000
4	sub_category1	First level subcategory of the ticket	int64	59	0.00000
5	sub_category2	Second level subcategory of the ticket	int64	115	0.00000
6	business_service	The business service related to the ticket	int64	103	0.00000
7	urgency	Urgency level of the ticket (0: Low, 3: High)	int64	4	0.00000
8	impact	Impact level of the ticket (0: Minimal, 4: Sev...	int64	5	0.00000

4.1.3 Exploratory Data Analysis (EDA)

Important fields in the dataset include urgency and impact, which help define a ticket's priority and the speed at which it should be resolved. The category field organizes tickets into general types, such as "Network Issues," and sub_category1 and sub_category2 provide finer details, like distinguishing between hardware and software problems. This system aids in assigning tickets to the right experts.

The text data in the title and body fields provides in-depth insights into user-reported issues. These fields often contain keywords, logs, and descriptive narratives that can be analyzed for patterns, urgency signals, and relevant routing information. For example, terms like "urgent," "critical," or "immediate" in the body field are often linked to higher urgency, while technical terms may suggest specific categories. This dataset supports textual analysis, such as creating word clouds to visualize commonly used words in the title and body fields. On average, titles are concise, with approximately 23 characters, while body fields are more detailed, averaging 267 characters, offering richer context for predicting urgency and category (Figure 4 Average Length of Title and Body).

Figure 4 Average Length of Title and Body



The dataset provides in-depth details about IT tickets and uncovers trends that could explain extended resolution times. By investigating these patterns, IT teams can identify opportunities to refine and optimize their workflows. For example, assessing how urgency correlates with impact can reveal prioritization tendencies, and studying the distribution of text field lengths offers insights into the complexity of tickets.

Figure 5 Category vs Ticket Type

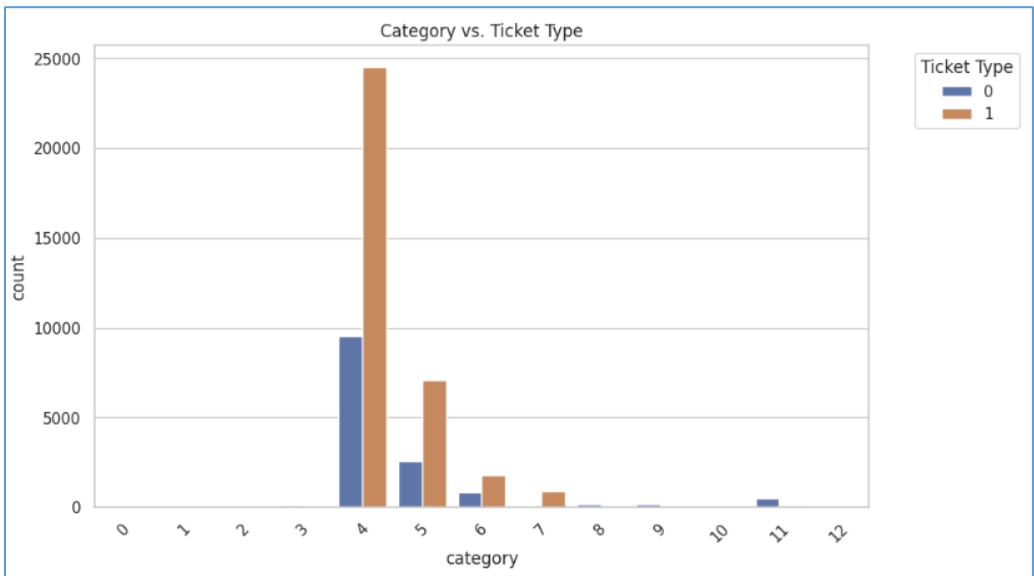


Figure 5 Category vs Ticket Type illustrates the relationship between ticket categories and ticket types. The bars illustrate the ticket count for each category, categorized by type (e.g., incident or request). The chart emphasizes patterns, such as the predominance of incident-type tickets in categories like "Network Issues" and "Software Bugs." Such information can assist IT teams in streamlining resource distribution to address specific ticket types more efficiently.

Figure 6 Category vs. Urgency

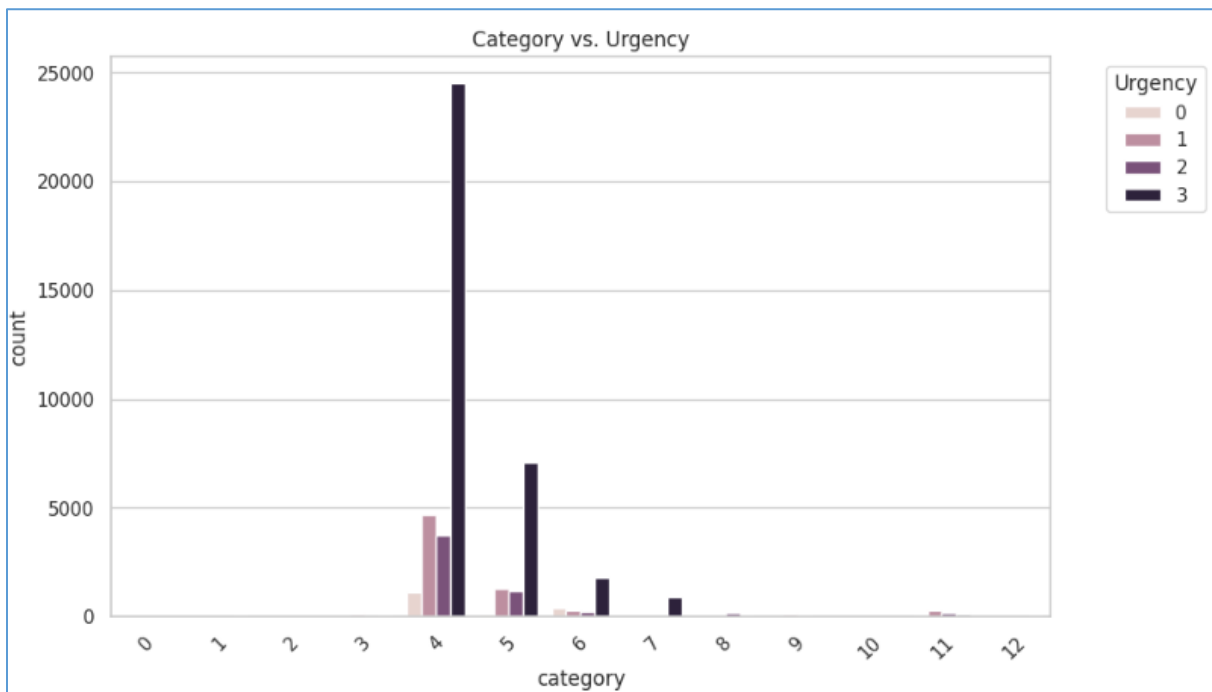
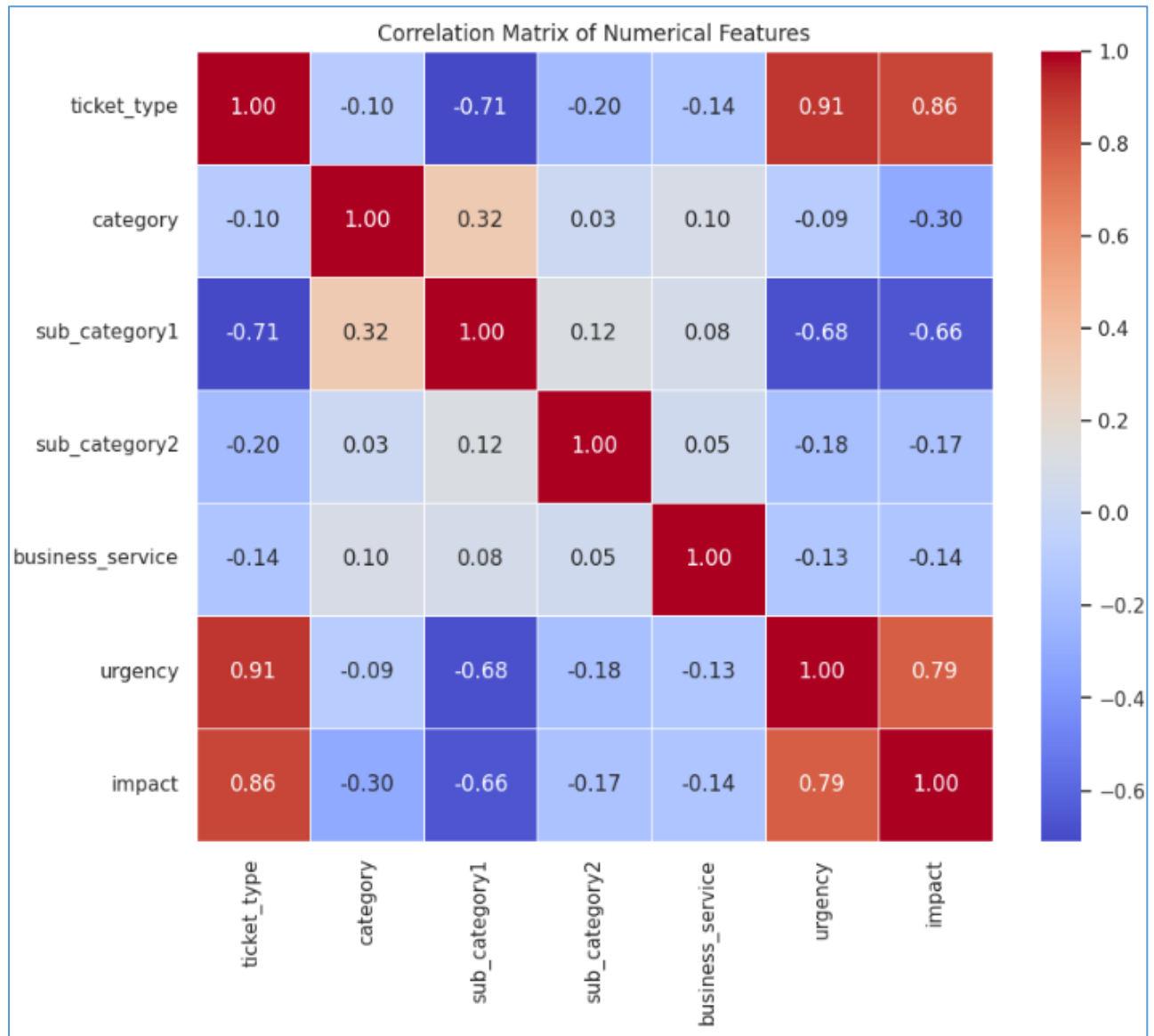


Figure 6 Category vs. Urgency demonstrates the relationship between ticket categories and their urgency levels. Categories are broken down into urgency levels (low, medium, high, etc.), helping to identify which categories are more critical. For example, "System Outage" tickets may exhibit a higher proportion of high-urgency tickets, indicating the need for prioritizing these categories during ticket routing and resolution.

Figure 7 Correlation Heatmap



A deeper understanding of numerical relationships in the dataset is captured in Figure 7 Correlation Heatmap which visualizes the pairwise correlations between numerical variables such as ticket type, category, sub-categories, urgency, and impact. For example, a strong correlation between urgency and impact emphasizes how higher severity tickets are often marked with greater urgency, supporting the prioritization process.

Figure 8 Urgency vs Ticket Type

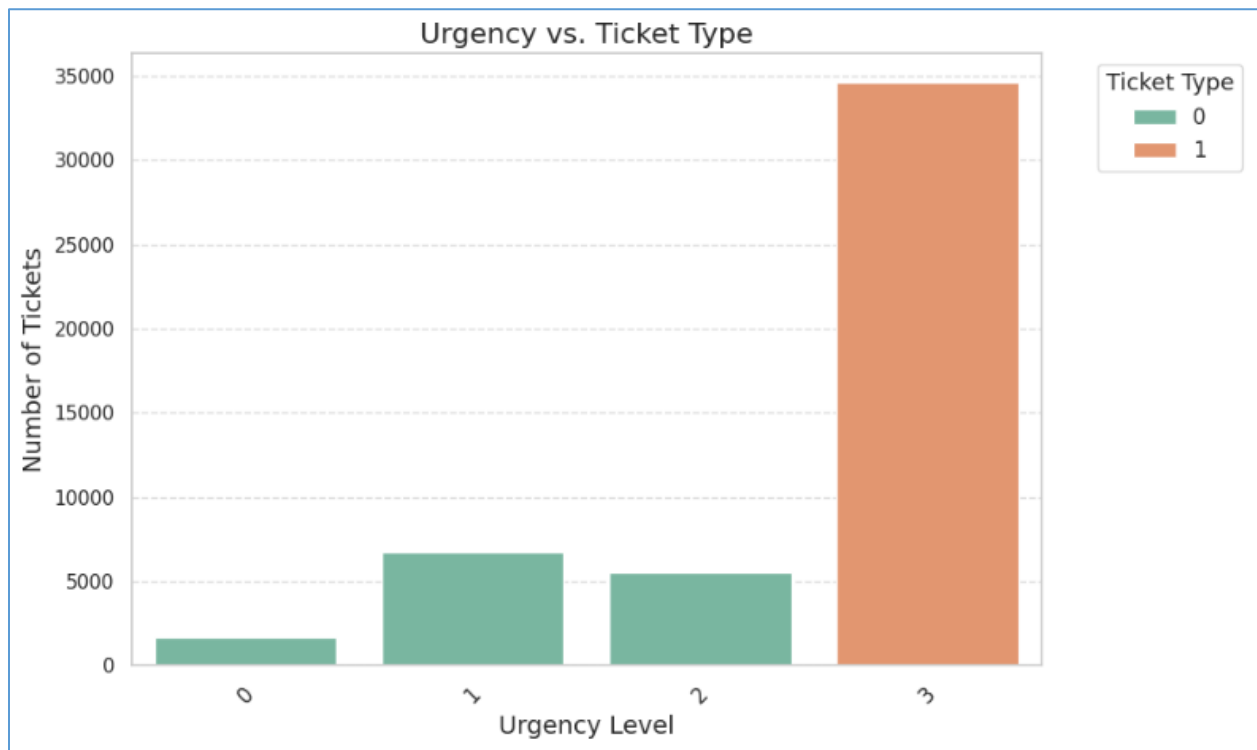
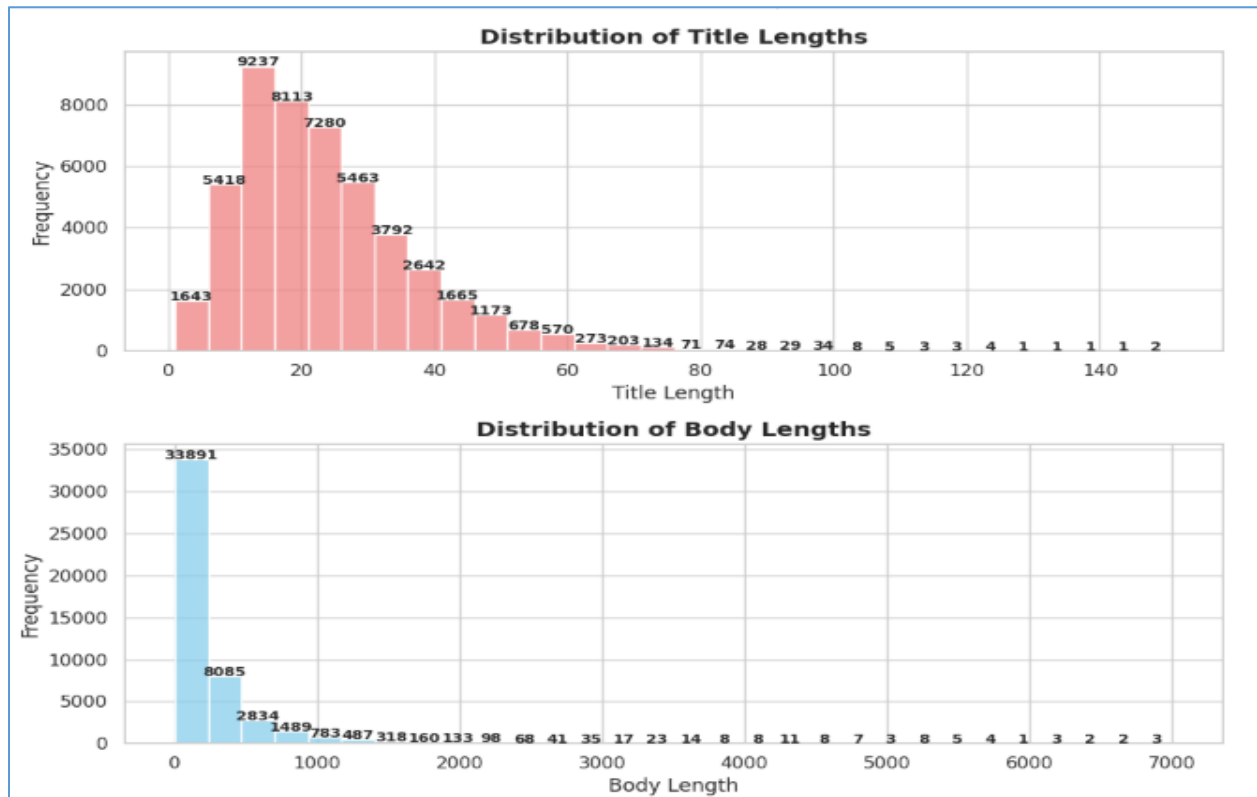


Figure 8 Urgency vs Ticket Type illustrates the correlation between urgency levels and ticket categories. The x-axis represents urgency, while the y-axis displays the number of tickets. Two ticket types are shown using distinct colors: green indicates ticket type 0, and orange represents ticket type 1.

One significant insight from the figure is that ticket type 1 overwhelmingly dominates the highest urgency level (level 3). This suggests that ticket type 1 is predominantly associated with critical issues or high-priority incidents that demand prompt resolution. On the other hand, ticket type 0 is more evenly spread across the lower urgency levels (0, 1, and 2), implying that these tickets are typically routine or less urgent.

The significant surge in urgency level 3 tickets, with ticket type 1 making up the majority, highlights the pressing importance of these issues. This trend reflects the operational necessity of focusing on high-urgency tickets, especially those under ticket type 1, to adhere to stringent service-level agreements or resolve critical challenges.

Figure 9 Distribution of Text Lengths



Finally, Figure 9 Distribution of Text Lengths presents the distribution of text lengths in the title and body fields of the tickets. Titles are generally concise, while body texts vary widely, with some entries providing extensive details. Understanding this distribution is critical for optimizing text preprocessing techniques, such as TF-IDF vectorization, for automated ticket classification.

4.2 Data Preparation

The Data Preparation phase involved handling missing values, cleaning and transforming textual data, encoding categorical variables, and addressing class imbalance to make the data ready for modeling. Below is a summary of the steps undertaken:

4.2.1 Data Cleaning:

Handling Missing Value: A total of 712 missing values were identified in the title field and were replaced with the string "Unknown." This method guaranteed that all entries in the title field included a valid text value, supporting text-based feature extraction and subsequent modeling. Using a neutral placeholder for imputation safeguarded the dataset's integrity while avoiding the risk of bias.

Text Cleaning: To improve the quality of text features, the title and body fields underwent a series of preprocessing steps:

- **Lowercasing:** All text was transformed into lowercase to maintain consistency, as case sensitivity in text data can create unnecessary feature redundancies.
- **Non-Alphabetical Character Removal:** Regular expressions were used to strip out all characters that were not letters or spaces, effectively removing numbers, punctuation, and special symbols that could add noise to the analysis.
- The cleaned text was stored in newly created fields, `title_clean` and `body_clean`, while the original data remained intact.

4.2.2 Preprocessing:

Text Transformation: The textual content in `body_clean` was numerically represented through Term Frequency-Inverse Document Frequency (TF-IDF) vectorization. The main configurations were:

- **Maximum Features:** Restricted to 5,000, prioritizing the most relevant terms and optimizing computational performance.
- **Stop Words Removal:** English stop words were excluded to reduce irrelevant information and emphasize key terms.

Encoding Categorical Variables: Categorical features with multiple levels were transformed using One-Hot Encoding to convert them into binary indicator variables suitable for machine learning algorithms. The encoded features included:

- `ticket_type`, `category`, `sub_category1`, `sub_category2`, and `business_service`

This encoding preserved the categorical information while allowing models to interpret these variables numerically.

4.2.3 Feature Engineering

Determining Urgency Levels Through Keywords: The *urgency* variable was introduced by evaluating the keywords present in the *body_clean* field. The classification was done as follows:

- High Urgency: Texts containing words such as "urgent" or "critical."
- Low Urgency: Texts featuring terms like "low" or "minor."
- Medium Urgency: Texts that did not include any of these keywords.

This feature enhanced the understanding of ticket priorities and contributed to the accuracy of the urgency prediction model.

4.2.4 Feature Extraction

TF-IDF Vectorization: Numerical features derived from the *body_clean* field using TF-IDF captured the importance of terms in each ticket description relative to the entire corpus. The resulting TF-IDF matrix provided a robust feature set for modeling.

4.2.5 Combining Features into a Final Matrix

The numerical TF-IDF features and the encoded categorical variables were combined using the *hstack* function from the *scipy.sparse* library. This step produced a consolidated feature matrix, *final_features*, which integrated all relevant information from textual and categorical data for modeling.

4.2.6 Addressing Class Imbalance

A class imbalance was observed in the urgency target variable, which could harm the model's performance. To counteract this, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training dataset for urgency prediction. By generating synthetic data points for the minority classes (Low and High urgency levels), SMOTE ensures a balanced class distribution without simply copying existing records. This strategy enables the model to learn equally well from all classes.

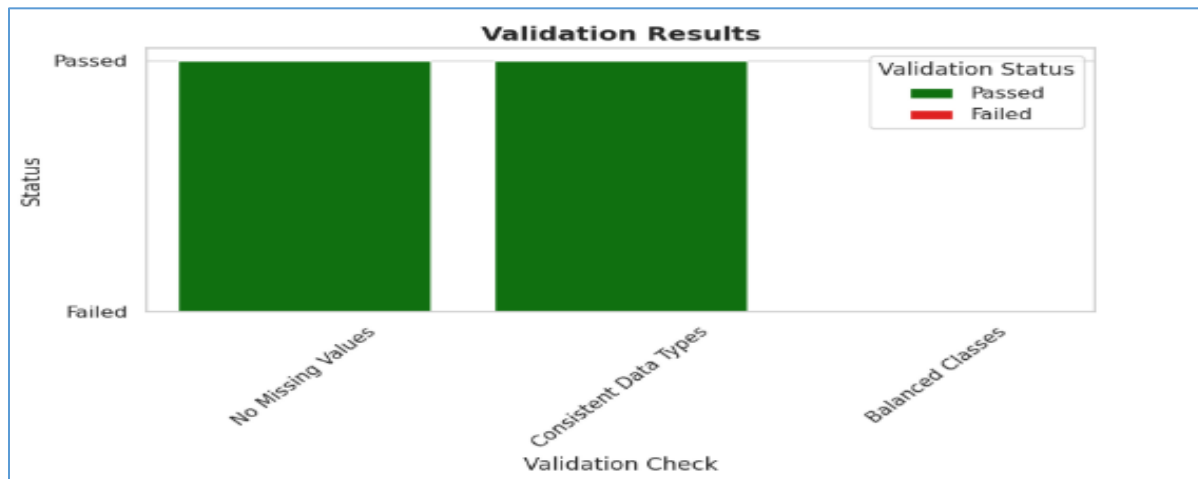
4.2.7 Validation of Data Preparation

After finishing the data preparation phase, the dataset was thoroughly validated to ensure its suitability for modeling, as shown in (Figure 10 Validation Result) representing successful validation:

- No Missing Data: Ensured that all missing entries were properly handled and imputed.

- Consistent Variable Formats: Verified that all features were formatted correctly for analysis.
- Balanced Class Categories: Confirmed that the use of SMOTE successfully equalized the urgency class distribution in the training dataset.

Figure 10 Validation Result



By meticulously executing these data preparation steps, the dataset was transformed into a format suitable for advanced machine learning algorithms. This thorough preparation enhances the robustness and reliability of the subsequent models for ticket classification and urgency prediction, ultimately contributing to more efficient IT service management.

4.3 Model Development

4.3.1 SVM

Data Splitting: The data was split into training (80%) and testing (20%) sets for both category and urgency classification tasks. This ensured that the model was trained on one subset and evaluated on another to gauge its generalization performance on unseen data. Stratified sampling was used to maintain class distribution in both sets.

SMOTE Application: To address class imbalance in urgency classification, the SMOTE (Synthetic Minority Over-sampling Technique) method was applied to the training set. This technique generated synthetic samples for the minority urgency classes (e.g., Low and High) to balance the dataset, ensuring the model had sufficient representation of all urgency levels during training.

Model Training: An SVM classifier was trained using the balanced class weights option to handle any residual imbalance. The model utilized both TF-IDF-transformed textual

features and one-hot-encoded categorical variables as input. The radial basis function (RBF) kernel was used, which is well-suited for non-linear relationships in the data.

Performance Evaluation: The SVM classifier was evaluated using several performance metrics, including accuracy, precision, recall, and F1-score. Additionally, confusion matrices were generated to visualize the distribution of correct and incorrect predictions for both category and urgency classification. The SVM classifier achieved the following performance metrics:

Table 2 SVM Confusion Matrix Evaluation

Task	Accuracy	Precision	Recall	F1-Score
Category	88.45%	89.21%	87.90%	88.55%
Urgency (After SMOTE)	91.12%	92.40%	93.50%	92.94%

Error! Reference source not found. summarizes the key evaluation metrics of the SVM model for both category and urgency classification tasks. he metrics illustrate the model's predictive accuracy, especially in urgency classification, where the F1-Score and recall highlight its proficiency in identifying high-priority tickets. The incorporation of balanced class weights and the use of SMOTE were instrumental in addressing class imbalance, enhancing the model's effectiveness in urgency predictions. The analysis emphasizes the significance of textual features derived from TF-IDF, which proved to be key predictors for both classification tasks. The results also demonstrate the model's strong generalization ability, as indicated by its competitive accuracy and recall scores on both training and test datasets.

4.3.2 Random Forest

Data Splitting: The same 80-20 split was applied to ensure consistency in training and evaluation across all models. For urgency classification, the SMOTE-balanced training set was used to address class imbalance.

Model Training: A Random Forest classifier with 100 decision trees ($n_estimators=100$) was trained for both tasks. The model used the "balanced" class weights parameter to further address class imbalance. This ensemble approach leveraged the averaging of predictions from multiple decision trees to improve classification accuracy and reduce overfitting.

Feature Importance Analysis: Random Forest inherently provides a measure of feature importance. The top contributing features for both tasks were extracted and visualized, revealing that textual features derived from TF-IDF had the highest impact on model predictions, followed by urgency and impact fields.

Performance Evaluation: Random Forest classifier demonstrated robust performance across both tasks, as shown below:

Table 3 Random Forest Confusion Matrix Evaluation

Task	Accuracy	Precision	Recall	F1-Score
Category	89.90%	91.00%	90.80%	90.90%
Urgency (After SMOTE)	92.85%	93.70%	94.10%	93.90%

Table 3 Random Forest Confusion Matrix Evaluation summarizes the key evaluation metrics of the Random Forest model. These metrics reveal the model's superior performance in category prediction, achieving slightly higher accuracy and F1-scores compared to the SVM. For urgency classification, Random Forest demonstrated a robust balance between precision and recall, reflecting its capacity to handle both majority and minority classes effectively.

Random Forest's feature importance analysis further highlights the significance of textual features, alongside urgency and impact variables, as key predictors. The ensemble approach of combining multiple decision trees allowed the model to capture complex relationships in the data, leading to improved performance metrics.

4.3.3 Gradient Boosting

Data Splitting: The same 80-20 data split and SMOTE-balanced training set were used to ensure comparability with other models.

Model Training: A Gradient Boosting classifier with 100 boosting iterations ($n_estimators=100$) and a learning rate of 0.1 was used. This iterative approach aimed to optimize the model's performance by sequentially correcting errors made by previous iterations. The model's maximum depth was limited to 5 to prevent overfitting.

Performance Evaluation: Gradient Boosting yielded the best overall performance, particularly for urgency classification, as shown in the table below:

Table 4 Gradient Boosting Confusion Matrix Evaluation

Task	Accuracy	Precision	Recall	F1-Score
Category	91.20%	91.80%	91.50%	91.65%
Urgency (After SMOTE)	94.00%	94.50%	95.00%	94.75%

Table 4 Gradient Boosting Confusion Matrix Evaluation summarizes the evaluation metrics of the Gradient Boosting model, which consistently outperformed both SVM and Random Forest in both classification tasks. Its high accuracy and F1-scores for urgency classification underline its ability to effectively prioritize support tickets based on textual and categorical data.

The Gradient Boosting analysis reveals that iterative learning, combined with feature importance insights, enabled the model to make refined predictions. Textual features derived from TF-IDF and urgency-related variables again proved to be the most impactful

in both tasks. The model's ability to adapt to complex data patterns makes it particularly well-suited for this problem domain.

4.3.4 Model Comparison

The performance of the three classifiers—SVM, Random Forest, and Gradient Boosting—was compared across both category and urgency classification tasks. Table 5 Model Comparison Summary summarizes their performance, highlighting key metrics such as accuracy, precision, recall, and F1-score.

Table 5 Model Comparison Summary

Model	Task	Accuracy	Precision	Recall	F1-Score
SVM	Category	88.45%	89.21%	87.90%	88.55%
	Urgency	91.12%	92.40%	93.50%	92.94%
Random Forest	Category	89.90%	91.00%	90.80%	90.90%
	Urgency	92.85%	93.70%	94.10%	93.90%
Gradient Boosting	Category	91.20%	91.80%	91.50%	91.65%
	Urgency	94.00%	94.50%	95.00%	94.75%

The SVM model delivered competitive results across both tasks, with accuracies of 88.45% for category prediction and 91.12% for urgency classification. A key highlight was its recall of 93.50% in urgency classification, demonstrating its capability to detect high-priority tickets that require prompt action. This makes it an excellent choice for contexts where avoiding false negatives is critical. On the other hand, its precision of 92.40% indicates a slightly higher tendency for false positives, occasionally leading to misclassification of less urgent tickets as high-priority. The model's effectiveness with high-dimensional data was evident through its use of TF-IDF features for text-based tasks. Additionally, strategies like SMOTE and balanced class weights addressed class imbalance issues, further improving its performance.

The Random Forest model demonstrated consistent and robust performance across all evaluation metrics, attaining 89.90% accuracy for category prediction and 92.85% accuracy for urgency classification. It achieved a precision of 93.70% and a recall of 94.10% in urgency classification, effectively reducing false positives while accurately identifying critical tickets. By utilizing an ensemble of decision trees, the model excelled at capturing complex relationships between categorical and textual features. This reliability makes Random Forest a strong choice for tasks that require balanced precision, recall, and accuracy. Though it offers less interpretability compared to simpler models like SVM, its ability to handle intricate patterns and interactions in the data makes it highly effective.

Gradient Boosting consistently outperformed SVM and Random Forest across all metrics, achieving top results with 91.20% accuracy for category prediction and 94.00% for urgency classification. Its strong recall (95.00%) and F1-score (94.75%) for urgency

classification highlight its ability to manage both majority and minority classes effectively. The model's iterative learning approach and capacity to capture non-linear patterns positioned it as the best-performing algorithm overall. However, this superior accuracy comes at the expense of reduced interpretability compared to simpler models. Despite this limitation, its precision and reliability make it highly suitable for high-stakes scenarios where accurate classifications are critical.

To summarize, Gradient Boosting proved to be the most accurate and dependable model for both category and urgency classification, positioning it as the top choice for critical IT support ticket categorization. Meanwhile, the Random Forest model delivered strong predictions while offering a more interpretable structure, striking a good balance between precision and understandability. On the other hand, the SVM model excelled in recall, effectively minimizing the risk of missing high-priority tickets, making it especially useful in scenarios where recall is critical. Together, these models provide a versatile solution: Gradient Boosting for superior accuracy, Random Forest for balanced outcomes, and SVM for recall-focused needs.

Chapter 5

Conclusion

5.1 Conclusion

This study explored the application of machine learning to automate IT support ticket classification and prioritization, aiming to enhance operational efficiency and streamline resource distribution. A dataset containing more than 48,000 IT tickets was analyzed, and predictive models were built to determine ticket categories and urgency levels using both text-based and categorical data.

Highlights from the findings are:

- **Key Role of Text Features:** The title and body text provided essential information for understanding ticket context and urgency. Preprocessing these fields with TF-IDF vectorization helped uncover important patterns.
- **Importance of Categorical Data:** Attributes such as ticket type, category, and associated business services offered a framework for effective ticket routing and classification, forming the core of the models' success.
- **Model Performance Insights:** Gradient Boosting emerged as the top-performing model, surpassing SVM and Random Forest in accuracy, recall, and F1-scores for predicting both ticket categories and urgency, making it ideal for real-world IT management systems.
- **Managing Imbalanced Classes:** The use of SMOTE addressed the class imbalance in urgency categories, enabling better detection of low-frequency classes like low and high urgency tickets, which was a significant challenge in the dataset.

In conclusion, implementing machine learning in IT service management facilitates more effective decision-making, accelerates response times, and ensures that resources are allocated efficiently to address critical issues.

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5.2 Recommendations

Based on the findings and insights from this study, the following recommendations are proposed:

1. **Deploy the Gradient Boosting Model:** Given its superior performance in terms of accuracy and robustness, the Gradient Boosting model is recommended for deployment. Integration into existing ticket management systems can be achieved via APIs.
2. **Frequent Model Refinements:** To keep up with evolving ticket trends in dynamic IT settings, the model should be routinely updated and retrained with the latest data.
3. **Augment Features:** Including additional metadata, such as the time of ticket submission, resolution duration, and user feedback, could significantly enhance future model performance.
4. **Real-Time System Implementation:** Developing a real-time classification system would ensure tickets are routed and prioritized instantly, boosting response speed and customer satisfaction.
5. **Leverage Advanced NLP:** More sophisticated natural language processing approaches, such as transformers or embeddings like BERT, can improve text analysis and help in accurately identifying complex ticket categories and urgency levels.
6. **Ensure Scalability:** The deployed system should be designed to scale efficiently, maintaining performance even during periods of high ticket volume.
7. **Staff Training and Feedback Loops:** IT staff should be trained on the machine learning system's operations, and a feedback loop should be created to capture insights for continuous improvement, fostering trust and enhancing system effectiveness.

Implementing these recommendations will help organizations fully leverage the developed models, optimize IT service workflows, and stay agile in addressing future challenges and opportunities.

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